

Instrument Rating

Flight Lesson Guide

An original flight teaching companion organized in the current publisher lesson sequence. Each flight begins with a complete operating plan, then assigns a realistic exercise budget and observable results. Ground entries point to the corresponding preparation; the detailed teaching is in the flight/device lessons. Source targets remain visible. Assessment and exercise selections must be adapted to the actual learner, aircraft, weather, and available facilities.

Sporty's Academy Sporty's Academy Instrument 141 TCO V18 / 06-17-2026; original Part 61 teaching layer. The source is a Part 141 TCO, not a Part 61 minimum-hour prescription.

Use this guide

Read the full teaching guide before flight. Use the compact checklist to sequence the flight. Minutes are pacing targets; repeat, stop, or defer an exercise when proficiency, conditions, or available time require it. Checkmarks indicate items covered, not a proficiency endorsement.

Aircraft values come from the applicable POH/AFM and approved checklist. Brief the aircraft-specific speeds, configurations, limitations, and any simulation method before conducting a maneuver.

Entry and preparation

- Confirm prerequisite certificates, privileges, medical eligibility where required, recency, and aircraft authorization.
- Use the applicable POH/AFM, current charts, approved equipment manuals, and current FAA standards.
- Reconcile the student's logbook against current Part 61 separately; completing these timed lessons does not establish certificate eligibility.
- Device time is not aircraft flight time. Credit only approved training/tasks under the device authorization and applicable regulation.

Sources and alignment

Sporty's Academy Instrument TCO - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

Current FAA Airman Certification Standards - Current FAA linked standard

Part 61 experience requirements - Verified 2026-10-10

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Instrument control and navigation

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- 18 · Stage I consolidation · 108 min dual
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Instrument control and navigation

Build reliable instrument control, backup control, radio/GPS navigation, and automation monitoring before instrument approaches.

Lesson 1 · Preparation: Altimeter, Types of Altitude, Vertical Speed Indicator

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

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Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 2 · Preparation: Instrument Scan, Instrument Interpretation, Aircraft Control

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

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Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 3 · Instrument attitude foundations

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72 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-59 Air 08-54	[] Build the scan and trim loop	Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents.
59-69 Air 54-64	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
69-72 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 3

Practice instrument attitude foundations. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Build the scan and trim loop · 46 minutes

Teach / coach: Start with a stable attitude and power setting. Demonstrate attitude change, cross-check of performance, small correction, then trim. Explain which indication confirms each change; let the learner verbalize the scan before removing prompts.

Student exercise: Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents. Lead the level-off, set cruise power, verify performance, and trim without abandoning the cross-check.

Practice: Two level-offs in each direction and at least two left/right turns; repeat the weakest transition.

Look for: Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.

Watch / correct: Fixation on one instrument - identify the missing cross-check and return to attitude/performance confirmation.

Watch / correct: Large chase corrections - stabilize attitude and power, wait for the response, then refine.

Stop / transition: Remove the view-limiting device or take control for unsafe trends, traffic conflict, disorientation, or an unrecoverable task load.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Build the scan and trim loop; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Instrument Takeoff. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 3 - source page 24 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 4 · Rates, turning transitions, and steep turns

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72 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-36 Air 08-31	[] Build the scan and trim loop	Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents.
36-59 Air 31-54	[] Rate changes and coordinated instrument turns	Fly a rate climb and descent, level off, then make climbing and descending turns.
59-69 Air 54-64	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
69-72 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 4

Practice rates, turning transitions, and steep turns. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.
- Anticipates level-offs and rollouts, keeps coordinated control, and identifies which correction fixes an altitude or airspeed trend.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Build the scan and trim loop · 23 minutes

Teach / coach: Start with a stable attitude and power setting. Demonstrate attitude change, cross-check of performance, small correction, then trim. Explain which indication confirms each change; let the learner verbalize the scan before removing prompts.

Student exercise: Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents. Lead the level-off, set cruise power, verify performance, and trim without abandoning the cross-check.

Practice: Two level-offs in each direction and at least two left/right turns; repeat the weakest transition.

Look for: Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.

Watch / correct: Fixation on one instrument - identify the missing cross-check and return to attitude/performance confirmation.

Watch / correct: Large chase corrections - stabilize attitude and power, wait for the response, then refine.

Stop / transition: Remove the view-limiting device or take control for unsafe trends, traffic conflict, disorientation, or an unrecoverable task load.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Build the scan and trim loop; current ACS and applicable POH/AFM

Rate changes and coordinated instrument turns · 23 minutes

Teach / coach: Compare a constant-airspeed climb with a constant-rate climb using the installed instruments. Demonstrate the scan expansion needed during climbing/descending turns and a steep turn; keep the learner from treating one instrument as the entire maneuver.

Student exercise: Fly a rate climb and descent, level off, then make climbing and descending turns. Finish with one steep turn each direction while monitoring load, attitude, altitude, and rollout.

Practice: One climb/descent pair and two turning transitions; one steep turn each way, with a corrective repeat.

Look for: Anticipates level-offs and rollouts, keeps coordinated control, and identifies which correction fixes an altitude or airspeed trend.

Watch / correct: Rollout late - lead the rollout and verify heading rather than increasing bank near the target.

Watch / correct: Altitude loss in steep turn - establish the appropriate attitude/power and widen the scan.

Stop / transition: Recover before exceeding the aircraft envelope or safety limits briefed for the session.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Rate changes and coordinated instrument turns; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.

- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Instrument Takeoff. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 4 - source page 25 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 5 · Instrument flight patterns

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72 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-36 Air 08-31	[] Combine attitude changes into flight patterns	Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates.
36-59 Air 31-54	[] Build the scan and trim loop	Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents.
59-69 Air 54-64	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
69-72 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 5

Practice instrument flight patterns. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.
- Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Combine attitude changes into flight patterns · 23 minutes

Teach / coach: Draw and brief a climb-turn-level-descent sequence before flight. First fly it with a spoken scan, then give only the next altitude/heading clearance. Use the pattern to expose transitions rather than add new maneuver complexity.

Student exercise: Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates. Call the next configuration and level-off in advance; repeat with fewer instructor cues.

Practice: Two complete patterns: coached, then independent; repeat one weak corner or level-off.

Look for: Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Watch / correct: Next leg surprises the learner - pause the sequence, restate the picture, then rehearse the transition.

Watch / correct: Trim omitted - stabilize each leg before moving to the next.

Stop / transition: Simplify to straight-and-level flight if the scan or orientation deteriorates.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Combine attitude changes into flight patterns; current ACS and applicable POH/AFM

Build the scan and trim loop · 23 minutes

Teach / coach: Start with a stable attitude and power setting. Demonstrate attitude change, cross-check of performance, small correction, then trim. Explain which indication confirms each change; let the learner verbalize the scan before removing prompts.

Student exercise: Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents. Lead the level-off, set cruise power, verify performance, and trim without abandoning the cross-check.

Practice: Two level-offs in each direction and at least two left/right turns; repeat the weakest transition.

Look for: Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.

Watch / correct: Fixation on one instrument - identify the missing cross-check and return to attitude/performance confirmation.

Watch / correct: Large chase corrections - stabilize attitude and power, wait for the response, then refine.

Stop / transition: Remove the view-limiting device or take control for unsafe trends, traffic conflict, disorientation, or an unrecoverable task load.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Build the scan and trim loop; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.

- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 5 - source page 26 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 6 · Preparation: Magnetic Compass Construction, Principles of Magnetic Attraction, Magnetic Dip

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation](#); assignments on the referenced TCO pages

[FAA aviation handbooks](#); use aircraft/equipment supplements too

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 7 · Compass, timed turns, and failures

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-47 Device	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
47-85 Device	[] Recognize and recover unusual attitudes	Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation...
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 7

Practice compass, timed turns, and failures. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Failure recognition and backup control · 39 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Recognize and recover unusual attitudes · 38 minutes

Teach / coach: Brief transfer of control and the approved recovery procedure. Present modest, realistic unusual attitudes within aircraft limitations; demonstrate how reliable indications distinguish nose-high, nose-low, spiral, and failed-display situations.

Student exercise: Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation or automation confusion caused the upset.

Practice: One nose-high and one nose-low recovery; repeat using the briefed reduced instrument set if the source lesson calls for it.

Look for: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Watch / correct: Control response before diagnosis - identify the attitude using reliable cues first.

Watch / correct: Recovery creates a secondary stall/overspeed - manage angle of attack, bank, power, and load according to the approved procedure.

Stop / transition: Instructor ends setup or takes control before aircraft limits, safety altitude, or recovery margin are threatened.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Recognize and recover unusual attitudes; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 7 - source page 28 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 8 · Backup control in aircraft

[Return to lesson index](#)

72 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-36 Air 08-31	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
36-59 Air 31-54	[] Combine attitude changes into flight patterns	Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates.
59-69 Air 54-64	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
69-72 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 8

Practice backup control in aircraft. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Failure recognition and backup control · 23 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Combine attitude changes into flight patterns · 23 minutes

Teach / coach: Draw and brief a climb-turn-level-descent sequence before flight. First fly it with a spoken scan, then give only the next altitude/heading clearance. Use the pattern to expose transitions rather than add new maneuver complexity.

Student exercise: Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates. Call the next configuration and level-off in advance; repeat with fewer instructor cues.

Practice: Two complete patterns: coached, then independent; repeat one weak corner or level-off.

Look for: Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Watch / correct: Next leg surprises the learner - pause the sequence, restate the picture, then rehearse the transition.

Watch / correct: Trim omitted - stabilize each leg before moving to the next.

Stop / transition: Simplify to straight-and-level flight if the scan or orientation deteriorates.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Combine attitude changes into flight patterns; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 8 - source page 29 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 9 · Unusual attitudes and backup patterns

[Return to lesson index](#)

72 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-29 Air 08-24	[] Recognize and recover unusual attitudes	Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation...
29-44 Air 24-39	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
44-59 Air 39-54	[] Combine attitude changes into flight patterns	Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates.
59-69 Air 54-64	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
69-72 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 9

Practice unusual attitudes and backup patterns. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Recognize and recover unusual attitudes · 16 minutes

Teach / coach: Brief transfer of control and the approved recovery procedure. Present modest, realistic unusual attitudes within aircraft limitations; demonstrate how reliable indications distinguish nose-high, nose-low, spiral, and failed-display situations.

Student exercise: Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation or automation confusion caused the upset.

Practice: One nose-high and one nose-low recovery; repeat using the briefed reduced instrument set if the source lesson calls for it.

Look for: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Watch / correct: Control response before diagnosis - identify the attitude using reliable cues first.

Watch / correct: Recovery creates a secondary stall/overspeed - manage angle of attack, bank, power, and load according to the approved procedure.

Stop / transition: Instructor ends setup or takes control before aircraft limits, safety altitude, or recovery margin are threatened.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Recognize and recover unusual attitudes; current ACS and applicable POH/AFM

Failure recognition and backup control · 15 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Combine attitude changes into flight patterns · 15 minutes

Teach / coach: Draw and brief a climb-turn-level-descent sequence before flight. First fly it with a spoken scan, then give only the next altitude/heading clearance. Use the pattern to expose transitions rather than add new maneuver complexity.

Student exercise: Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates. Call the next configuration and level-off in advance; repeat with fewer instructor cues.

Practice: Two complete patterns: coached, then independent; repeat one weak corner or level-off.

Look for: Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Watch / correct: Next leg surprises the learner - pause the sequence, restate the picture, then rehearse the transition.

Watch / correct: Trim omitted - stabilize each leg before moving to the next.

Stop / transition: Simplify to straight-and-level flight if the scan or orientation deteriorates.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Combine attitude changes into flight patterns; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 9 - source page 30 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 10 · Preparation: VOR Principles of Operation/ Transmitter/ Receiver, VOR Class Designations & Service Volumes, ...

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 11 · VOR navigation in the device

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-85 Device	[] VOR orientation, intercept, and wind correction	Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound.
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 11

Practice vor navigation in the device. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

VOR orientation, intercept, and wind correction · 77 minutes

Teach / coach: On the ground confirm receiver-check status, tune and identify the facility, and draw the selected radial/course relationship. Demonstrate one intercept with the correct source and TO/FROM interpretation, then compare tracking corrections before and after wind change.

Student exercise: Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound. Detect station passage, change the course, and cross-check position with another valid source.

Practice: Two intercepts with opposite geometry, inbound/outbound tracking, and one station passage or device equivalent.

Look for: Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.

Watch / correct: Reverse sensing - verify selected course, TO/FROM indication, and intended direction.

Watch / correct: Intercept mistaken for track - reduce to a sustainable wind correction after capture.

Stop / transition: Discontinue use of an unreliable or unidentifiable signal; use a briefed legal alternative and tell ATC when required.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; VOR orientation, intercept, and wind correction; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 11 - source page 32 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 12 · VOR tracking in flight

[Return to lesson index](#)

72 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-36 Air 08-31	[] VOR orientation, intercept, and wind correction	Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound.
36-59 Air 31-54	[] Combine attitude changes into flight patterns	Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates.
59-69 Air 54-64	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
69-72 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 12

Practice vor tracking in flight. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.
- Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

VOR orientation, intercept, and wind correction · 23 minutes

Teach / coach: On the ground confirm receiver-check status, tune and identify the facility, and draw the selected radial/course relationship. Demonstrate one intercept with the correct source and TO/FROM interpretation, then compare tracking corrections before and after wind change.

Student exercise: Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound. Detect station passage, change the course, and cross-check position with another valid source.

Practice: Two intercepts with opposite geometry, inbound/outbound tracking, and one station passage or device equivalent.

Look for: Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.

Watch / correct: Reverse sensing - verify selected course, TO/FROM indication, and intended direction.

Watch / correct: Intercept mistaken for track - reduce to a sustainable wind correction after capture.

Stop / transition: Discontinue use of an unreliable or unidentifiable signal; use a briefed legal alternative and tell ATC when required.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; VOR orientation, intercept, and wind correction; current ACS and applicable POH/AFM

Combine attitude changes into flight patterns · 23 minutes

Teach / coach: Draw and brief a climb-turn-level-descent sequence before flight. First fly it with a spoken scan, then give only the next altitude/heading clearance. Use the pattern to expose transitions rather than add new maneuver complexity.

Student exercise: Fly the briefed rectangular or vertical pattern using assigned headings, altitudes, and rates. Call the next configuration and level-off in advance; repeat with fewer instructor cues.

Practice: Two complete patterns: coached, then independent; repeat one weak corner or level-off.

Look for: Moves smoothly between maneuvers, stays oriented, and maintains the selected parameters without losing the cross-check.

Watch / correct: Next leg surprises the learner - pause the sequence, restate the picture, then rehearse the transition.

Watch / correct: Trim omitted - stabilize each leg before moving to the next.

Stop / transition: Simplify to straight-and-level flight if the scan or orientation deteriorates.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Combine attitude changes into flight patterns; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 12 - source page 33 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 13 · Preparation: GPS Use Under IFR, Substitution of GPS for Other Navigation Radios Under IFR, GPS Principles o...

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 14 · Installed GPS workflow

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-85 Device	[] GPS setup, sequencing, and integrity	Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest...
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 14

Practice installed gps workflow. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Verifies every clearance and active leg, understands available guidance, and preserves control while programming.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

GPS setup, sequencing, and integrity · 77 minutes

Teach / coach: Use the installed navigator manual to show source selection, flight-plan loading, leg activation, direct-to, nearest, and scaling changes. Explain integrity alerts and how WAAS/RAIM availability affects a planned procedure; confirm the next mode before changing it.

Student exercise: Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest alternate. State navigation source, active leg, scaling, and response to an integrity alert.

Practice: One complete route plus one reroute and one integrity/failure decision.

Look for: Verifies every clearance and active leg, understands available guidance, and preserves control while programming.

Watch / correct: Wrong leg activated - compare clearance, map, flight-plan list, and CDI before executing.

Watch / correct: Heads-down programming - stabilize first; ask for delay or use a simpler navigation method.

Stop / transition: Stop programming for control/traffic priorities; discontinue approach guidance when integrity or required equipment is unavailable.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; GPS setup, sequencing, and integrity; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 14 - source page 35 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 15 · Preparation: Autopilot Principles of Operation, Autopilot Errors and Irregularities, Autopilot Limitations

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 16 · GPS and autopilot integration

[Return to lesson index](#)

108 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-41 Air 08-36	[] GPS setup, sequencing, and integrity	Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest...
41-68 Air 36-63	[] Autopilot modes and manual reversion	Select heading, navigation, altitude, and permitted climb/descent modes.
68-95 Air 63-90	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
95-105 Air 90-100	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
105-108 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 16

Practice gps and autopilot integration. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Verifies every clearance and active leg, understands available guidance, and preserves control while programming.
- Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

GPS setup, sequencing, and integrity · 28 minutes

Teach / coach: Use the installed navigator manual to show source selection, flight-plan loading, leg activation, direct-to, nearest, and scaling changes. Explain integrity alerts and how WAAS/RAIM availability affects a planned procedure; confirm the next mode before changing it.

Student exercise: Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest alternate. State navigation source, active leg, scaling, and response to an integrity alert.

Practice: One complete route plus one reroute and one integrity/failure decision.

Look for: Verifies every clearance and active leg, understands available guidance, and preserves control while programming.

Watch / correct: Wrong leg activated - compare clearance, map, flight-plan list, and CDI before executing.

Watch / correct: Heads-down programming - stabilize first; ask for delay or use a simpler navigation method.

Stop / transition: Stop programming for control/traffic priorities; discontinue approach guidance when integrity or required equipment is unavailable.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; GPS setup, sequencing, and integrity; current ACS and applicable POH/AFM

Autopilot modes and manual reversion · 27 minutes

Teach / coach: Brief the approved engagement envelope, limitations, disconnect methods, and failure signs. Demonstrate each installed mode by stating what is armed, active, commanded, and actually happening; then disconnect and hand-fly the same segment.

Student exercise: Select heading, navigation, altitude, and permitted climb/descent modes. Confirm annunciations and resulting flight path after every change. Recover from a simulated wrong mode or failed automation using the approved disconnect and manual control.

Practice: One mode sequence, one capture verification, and two different disconnect/reversion methods if installed.

Look for: Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.

Watch / correct: Armed mistaken for active - point to annunciations and confirm actual capture.

Watch / correct: Fighting engaged servos - use the approved disconnect and stabilize manually.

Stop / transition: Disengage for unexpected flight path, unresolved mode confusion, or operation outside approved limitations.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Autopilot modes and manual reversion; current ACS and applicable POH/AFM

Failure recognition and backup control · 27 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: VOR Procedures (if equipped). Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 16 - source page 37 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 17 · Preparation: 14 CFR Part 1, 14 CFR Part 43, 14 CFR Part 61

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 18 · Stage I consolidation

[Return to lesson index](#)

108 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-30 Air 08-25	[] Rate changes and coordinated instrument turns	Fly a rate climb and descent, level off, then make climbing and descending turns.
30-47 Air 25-42	[] VOR orientation, intercept, and wind correction	Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound.
47-63 Air 42-58	[] GPS setup, sequencing, and integrity	Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest...
63-79 Air 58-74	[] Autopilot modes and manual reversion	Select heading, navigation, altitude, and permitted climb/descent modes.
79-95 Air 74-90	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
95-105 Air 90-100	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
105-108 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 18

Practice stage i consolidation. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Anticipates level-offs and rollouts, keeps coordinated control, and identifies which correction fixes an altitude or airspeed trend.
- Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.
- Verifies every clearance and active leg, understands available guidance, and preserves control while programming.
- Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Rate changes and coordinated instrument turns · 17 minutes

Teach / coach: Compare a constant-airspeed climb with a constant-rate climb using the installed instruments. Demonstrate the scan expansion needed during climbing/descending turns and a steep turn; keep the learner from treating one instrument as the entire maneuver.

Student exercise: Fly a rate climb and descent, level off, then make climbing and descending turns. Finish with one steep turn each direction while monitoring load, attitude, altitude, and rollout.

Practice: One climb/descent pair and two turning transitions; one steep turn each way, with a corrective repeat.

Look for: Anticipates level-offs and rollouts, keeps coordinated control, and identifies which correction fixes an altitude or airspeed trend.

Watch / correct: Rollout late - lead the rollout and verify heading rather than increasing bank near the target.

Watch / correct: Altitude loss in steep turn - establish the appropriate attitude/power and widen the scan.

Stop / transition: Recover before exceeding the aircraft envelope or safety limits briefed for the session.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Rate changes and coordinated instrument turns; current ACS and applicable POH/AFM

VOR orientation, intercept, and wind correction · 17 minutes

Teach / coach: On the ground confirm receiver-check status, tune and identify the facility, and draw the selected radial/course relationship. Demonstrate one intercept with the correct source and TO/FROM interpretation, then compare tracking corrections before and after wind change.

Student exercise: Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound. Detect station passage, change the course, and cross-check position with another valid source.

Practice: Two intercepts with opposite geometry, inbound/outbound tracking, and one station passage or device equivalent.

Look for: Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.

Watch / correct: Reverse sensing - verify selected course, TO/FROM indication, and intended direction.

Watch / correct: Intercept mistaken for track - reduce to a sustainable wind correction after capture.

Stop / transition: Discontinue use of an unreliable or unidentifiable signal; use a briefed legal alternative and tell ATC when required.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; VOR orientation, intercept, and wind correction; current ACS and applicable POH/AFM

GPS setup, sequencing, and integrity · 16 minutes

Teach / coach: Use the installed navigator manual to show source selection, flight-plan loading, leg activation, direct-to, nearest, and scaling changes. Explain integrity alerts and how WAAS/RAIM availability affects a planned procedure; confirm the next mode before changing it.

Student exercise: Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest alternate. State navigation source, active leg, scaling, and response to an integrity alert.

Practice: One complete route plus one reroute and one integrity/failure decision.

Look for: Verifies every clearance and active leg, understands available guidance, and preserves control while programming.

Watch / correct: Wrong leg activated - compare clearance, map, flight-plan list, and CDI before executing.

Watch / correct: Heads-down programming - stabilize first; ask for delay or use a simpler navigation method.

Stop / transition: Stop programming for control/traffic priorities; discontinue approach guidance when integrity or required equipment is unavailable.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; GPS setup, sequencing, and integrity; current ACS and applicable POH/AFM

Autopilot modes and manual reversion · 16 minutes

Teach / coach: Brief the approved engagement envelope, limitations, disconnect methods, and failure signs. Demonstrate each installed mode by stating what is armed, active, commanded, and actually happening; then disconnect and hand-fly the same segment.

Student exercise: Select heading, navigation, altitude, and permitted climb/descent modes. Confirm annunciations and resulting flight path after every change. Recover from a simulated wrong mode or failed automation using the approved disconnect and manual control.

Practice: One mode sequence, one capture verification, and two different disconnect/reversion methods if installed.

Look for: Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.

Watch / correct: Armed mistaken for active - point to annunciations and confirm actual capture.

Watch / correct: Fighting engaged servos - use the approved disconnect and stabilize manually.

Stop / transition: Disengage for unexpected flight path, unresolved mode confusion, or operation outside approved limitations.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Autopilot modes and manual reversion; current ACS and applicable POH/AFM

Failure recognition and backup control · 16 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief - 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Instrument Takeoff. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 18 - source page 39 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 19 · Stage I assessment

[Return to lesson index](#)

84 min check | Brief 39 | Preflight 15 | Debrief 21

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-25 Air 08-20	[] Build the scan and trim loop	Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents.
25-37 Air 20-32	[] Recognize and recover unusual attitudes	Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation...
37-49 Air 32-44	[] VOR orientation, intercept, and wind correction	Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound.
49-60 Air 44-55	[] GPS setup, sequencing, and integrity	Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest...
60-71 Air 55-66	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
71-81 Air 66-76	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
81-84 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 19

Assess independent performance and decisions using current FAA criteria; pause coaching except to protect safety.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.
- Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.
- Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.
- Verifies every clearance and active leg, understands available guidance, and preserves control while programming.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 39 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Assess the learner's departure, configuration, and course intercept independently; give no coaching during the check.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Build the scan and trim loop · 12 minutes

Teach / coach: Assign the briefed build the scan and trim loop scenario and observe setup, execution, and decisions without coaching. Assess this result: Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Establish straight-and-level flight, make assigned standard-rate turns, then constant-airspeed climbs and descents. Lead the level-off, set cruise power, verify performance, and trim without abandoning the cross-check.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Maintains an effective cross-check and corrects trends before large deviations; uses applicable ACS tolerances at the evaluation stage.

Watch / correct: Fixation on one instrument - identify the missing cross-check and return to attitude/performance confirmation.

Watch / correct: Large chase corrections - stabilize attitude and power, wait for the response, then refine.

Stop / transition: Remove the view-limiting device or take control for unsafe trends, traffic conflict, disorientation, or an unrecoverable task load.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Build the scan and trim loop; current ACS and applicable POH/AFM

Recognize and recover unusual attitudes · 12 minutes

Teach / coach: Assign the briefed recognize and recover unusual attitudes scenario and observe setup, execution, and decisions without coaching. Assess this result: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation or automation confusion caused the upset.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Watch / correct: Control response before diagnosis - identify the attitude using reliable cues first.

Watch / correct: Recovery creates a secondary stall/overspeed - manage angle of attack, bank, power, and load according to the approved procedure.

Stop / transition: Instructor ends setup or takes control before aircraft limits, safety altitude, or recovery margin are threatened.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Recognize and recover unusual attitudes; current ACS and applicable POH/AFM

VOR orientation, intercept, and wind correction · 12 minutes

Teach / coach: Assign the briefed vor orientation, intercept, and wind correction scenario and observe setup, execution, and decisions without coaching. Assess this result: Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Tune, identify, select the course, state position relative to the station, intercept, and track inbound and outbound. Detect station passage, change the course, and cross-check position with another valid source.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Uses a valid identified signal, distinguishes radial from course, and corrects wind without chasing the needle.

Watch / correct: Reverse sensing - verify selected course, TO/FROM indication, and intended direction.

Watch / correct: Intercept mistaken for track - reduce to a sustainable wind correction after capture.

Stop / transition: Discontinue use of an unreliable or unidentifiable signal; use a briefed legal alternative and tell ATC when required.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; VOR orientation, intercept, and wind correction; current ACS and applicable POH/AFM

GPS setup, sequencing, and integrity · 11 minutes

Teach / coach: Assign the briefed gps setup, sequencing, and integrity scenario and observe setup, execution, and decisions without coaching. Assess this result: Verifies every clearance and active leg, understands available guidance, and preserves control while programming. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Load and verify a short route, intercept an active leg, observe waypoint sequencing, accept a reroute with direct-to, then choose an appropriate nearest alternate. State navigation source, active leg,

scaling, and response to an integrity alert.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Verifies every clearance and active leg, understands available guidance, and preserves control while programming.

Watch / correct: Wrong leg activated - compare clearance, map, flight-plan list, and CDI before executing.

Watch / correct: Heads-down programming - stabilize first; ask for delay or use a simpler navigation method.

Stop / transition: Stop programming for control/traffic priorities; discontinue approach guidance when integrity or required equipment is unavailable.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; GPS setup, sequencing, and integrity; current ACS and applicable POH/AFM

Failure recognition and backup control · 11 minutes

Teach / coach: Assign the briefed failure recognition and backup control scenario and observe setup, execution, and decisions without coaching. Assess this result: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 21 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Instrument Takeoff. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Steep Turns. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Autopilot Procedures (if equipped). Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 19 - source page 41, 42 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Approaches and terminal procedures

Combine clearances, holding, lateral/vertical approaches, missed approaches, degraded equipment, and landing decisions.

Lesson 20 · Preparation: Purpose of Holding, Holding Airspace, Holding Pattern Legs

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 21 · Holding and clearance practice

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-85 Device	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 21

Practice holding and clearance practice. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Clearance, holding entry, and wind correction · 77 minutes

Teach / coach: Have the learner read back and sketch the clearance, fix, course, direction, restriction, and expect-further-clearance information. Demonstrate fix-crossing flow and one entry; explain that the chosen entry must remain within protected airspace.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: Two entries from different arrival directions and two complete circuits; one timed and one distance-based hold when available.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Timing during Holds. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 21 - source page 44 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 22 · Preparation: Terminal Procedures Publication, Aircraft Approach Categories, Inoperative Components or Visua...

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 23 - Preparation: Terminal Arrival Area (TAA) Approaches, Vectored Approaches, Non-precision Instrument Approaches

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 24 · Lateral approaches and missed approaches

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-34 Device	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
34-60 Device	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
60-85 Device	[] IFR route, weather, and contingency decisions	File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg.
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 24

Practice lateral approaches and missed approaches. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Clearance, holding entry, and wind correction · 26 minutes

Teach / coach: Have the learner read back and sketch the clearance, fix, course, direction, restriction, and expect-further-clearance information. Demonstrate fix-crossing flow and one entry; explain that the chosen entry must remain within protected airspace.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: Two entries from different arrival directions and two complete circuits; one timed and one distance-based hold when available.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Lateral approach and missed approach · 26 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

IFR route, weather, and contingency decisions · 25 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg. Select an alternate before the escape option disappears; explain the applicable lost-communications response and request assistance while communications work.

Practice: At least two route checkpoints, one changed-clearance readback, one diversion decision, and approaches at planned destinations.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Departure Vectors to Filed Route. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Approach Setup and Briefing. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from an Approach. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 24 - source page 47 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 25 · Preparation: Clearance Delivery, Ground Control, Tower Control (Local Control)

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 26 · Aircraft IFR departure and lateral approaches

[Return to lesson index](#)

108 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-41 Air 08-36	[] IFR route, weather, and contingency decisions	File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg.
41-68 Air 36-63	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
68-95 Air 63-90	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
95-105 Air 90-100	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
105-108 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 26

Practice aircraft ifr departure and lateral approaches. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.
- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

IFR route, weather, and contingency decisions · 28 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg. Select an alternate before the escape option disappears; explain the applicable lost-communications response and request assistance while communications work.

Practice: At least two route checkpoints, one changed-clearance readback, one diversion decision, and approaches at planned destinations.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

Clearance, holding entry, and wind correction · 27 minutes

Teach / coach: Have the learner read back and sketch the clearance, fix, course, direction, restriction, and expect-further-clearance information. Demonstrate fix-crossing flow and one entry; explain that the chosen entry must remain within protected airspace.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: Two entries from different arrival directions and two complete circuits; one timed and one distance-based hold when available.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Lateral approach and missed approach · 27 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Departure Vectors to Filed Route. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Pilot Navigation to Filed Route. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Approach Setup and Briefing. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 26 - source page 49 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 27 · Preparation: Air Traffic Clearance, Contact Approach, Visual Approach

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 28 · Preparation: Localizer Principles of Operation, Glideslope Principles of Operation, Marker Beacons

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 29 · ILS and vertical guidance practice

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-47 Device	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
47-85 Device	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 29

Practice ils and vertical guidance practice. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Source clarification: Source task row prints LP within APV. LP is lateral-only; this teaching plan uses LPV or LNAV/VNAV for approved vertical guidance and teaches LP with lateral-only minima.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

ILS and approved vertical guidance · 39 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Lateral approach and missed approach · 38 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief - 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 29 - source page 52 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 30 · Aircraft vertical and lateral approaches

[Return to lesson index](#)

120 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-45 Air 08-40	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
45-76 Air 40-71	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
76-107 Air 71-102	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 30

Practice aircraft vertical and lateral approaches. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Clearance, holding entry, and wind correction · 32 minutes

Teach / coach: Have the learner read back and sketch the clearance, fix, course, direction, restriction, and expect-further-clearance information. Demonstrate fix-crossing flow and one entry; explain that the chosen entry must remain within protected airspace.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: Two entries from different arrival directions and two complete circuits; one timed and one distance-based hold when available.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 31 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Lateral approach and missed approach · 31 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Filing an IFR Flight Plan. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: IFR Navigation. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Approach Setup and Brief. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 30 - source page 53 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 31 · Preparation: Autopilot Approach Operations and Limitations, Non-precision Approaches with Autopilot, Preci...

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 32 · Degraded approaches, arc, and circle

[Return to lesson index](#)

90 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-24 Device	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
24-40 Device	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
40-55 Device	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
55-70 Device	[] DME arc and position cross-check	Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead...
70-85 Device	[] Visual transition and circling decision	Maintain required visual contact and circling altitude while positioning for a stabilized landing.
85-90 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 32

Practice degraded approaches, arc, and circle. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Source clarification: The lesson heading prints Stage I; the course section/sequence places this lesson in Stage II. Navigation uses the surrounding course section.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.
- Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Failure recognition and backup control · 16 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Lateral approach and missed approach · 16 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 15 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

DME arc and position cross-check · 15 minutes

Teach / coach: Brief the charted arc, altitude, direction, lead points, and approved distance source.

Demonstrate a small heading change to maintain distance and anticipation of the inbound course; explain slant-range and equipment substitution limitations.

Student exercise: Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead the inbound-course intercept.

Practice: One arc each direction in the device; aircraft session one published arc plus a repeat of the intercept if time permits.

Look for: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.

Watch / correct: Late arc intercept - calculate/brief lead using speed and geometry before the turn.

Watch / correct: Large distance chase - make small heading changes and reassess trend.

Stop / transition: Ask for vectors or another procedure if equipment, clearance, or containment is uncertain.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; DME arc and position cross-check; current ACS and applicable POH/AFM

Visual transition and circling decision · 15 minutes

Teach / coach: Brief approach category, published circling area, wind, visual references, missed procedure, and aircraft/school operating limits. Demonstrate the ground track and decision gates on a chart before using an authorized visual or device scenario.

Student exercise: Maintain required visual contact and circling altitude while positioning for a stabilized landing. Reject a poor position early; explain and fly the appropriate missed procedure if visual contact or a safe landing path is lost.

Practice: One straight-in visual transition and one authorized circling scenario; the device can rehearse loss of visual reference.

Look for: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Watch / correct: Tightening bank to rescue the landing - abandon the unstable position and go missed.

Watch / correct: Visual contact lost with no plan - brief the missed transition before starting the circle.

Stop / transition: Go missed immediately when required references, protected-area assurance, or a stabilized landing path are lost.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Visual transition and circling decision; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Inoperative Components or Visual Aids Table. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 32 - source page 55 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 33 · Aircraft degraded approach decisions

[Return to lesson index](#)

120 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-37 Air 08-32	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
37-61 Air 32-56	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
61-84 Air 56-79	[] DME arc and position cross-check	Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead...
84-107 Air 79-102	[] Visual transition and circling decision	Maintain required visual contact and circling altitude while positioning for a stabilized landing.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 33

Practice aircraft degraded approach decisions. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.
- Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Failure recognition and backup control · 24 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Lateral approach and missed approach · 24 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

DME arc and position cross-check · 23 minutes

Teach / coach: Brief the charted arc, altitude, direction, lead points, and approved distance source.

Demonstrate a small heading change to maintain distance and anticipation of the inbound course; explain slant-range and equipment substitution limitations.

Student exercise: Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead the inbound-course intercept.

Practice: One arc each direction in the device; aircraft session one published arc plus a repeat of the intercept if time permits.

Look for: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.

Watch / correct: Late arc intercept - calculate/brief lead using speed and geometry before the turn.

Watch / correct: Large distance chase - make small heading changes and reassess trend.

Stop / transition: Ask for vectors or another procedure if equipment, clearance, or containment is uncertain.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; DME arc and position cross-check; current ACS and applicable POH/AFM

Visual transition and circling decision · 23 minutes

Teach / coach: Brief approach category, published circling area, wind, visual references, missed procedure, and aircraft/school operating limits. **Demonstrate** the ground track and decision gates on a chart before using an authorized visual or device scenario.

Student exercise: Maintain required visual contact and circling altitude while positioning for a stabilized landing. Reject a poor position early; explain and fly the appropriate missed procedure if visual contact or a safe landing path is lost.

Practice: One straight-in visual transition and one authorized circling scenario; the device can rehearse loss of visual reference.

Look for: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Watch / correct: Tightening bank to rescue the landing - abandon the unstable position and go missed.

Watch / correct: Visual contact lost with no plan - brief the missed transition before starting the circle.

Stop / transition: Go missed immediately when required references, protected-area assurance, or a stabilized landing path are lost.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Visual transition and circling decision; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 33 - source page 56 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 34 - Preparation: Required Conditions for Ice Formation, Formation of Frost, Formation of Clear Ice

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 35 · Preparation: Conditions Required for Thunderstorms, Thunderstorm Lifecycle, Air Mass Thunderstorms

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 36 · Coupled approach and missed workflow

[Return to lesson index](#)

120 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-32 Air 08-27	[] Autopilot modes and manual reversion	Select heading, navigation, altitude, and permitted climb/descent modes.
32-51 Air 27-46	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
51-70 Air 46-65	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
70-89 Air 65-84	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
89-107 Air 84-102	[] Visual transition and circling decision	Maintain required visual contact and circling altitude while positioning for a stabilized landing.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 36

Practice coupled approach and missed workflow. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Autopilot modes and manual reversion · 19 minutes

Teach / coach: Brief the approved engagement envelope, limitations, disconnect methods, and failure signs. Demonstrate each installed mode by stating what is armed, active, commanded, and actually happening; then disconnect and hand-fly the same segment.

Student exercise: Select heading, navigation, altitude, and permitted climb/descent modes. Confirm annunciations and resulting flight path after every change. Recover from a simulated wrong mode or failed automation using the approved disconnect and manual control.

Practice: One mode sequence, one capture verification, and two different disconnect/reversion methods if installed.

Look for: Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.

Watch / correct: Armed mistaken for active - point to annunciations and confirm actual capture.

Watch / correct: Fighting engaged servos - use the approved disconnect and stabilize manually.

Stop / transition: Disengage for unexpected flight path, unresolved mode confusion, or operation outside approved limitations.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Autopilot modes and manual reversion; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 19 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Lateral approach and missed approach · 19 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

Clearance, holding entry, and wind correction · 19 minutes

Teach / coach: Have the learner read back and sketch the clearance, fix, course, direction, restriction, and expect-further-clearance information. Demonstrate fix-crossing flow and one entry; explain that the chosen entry must remain within protected airspace.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: Two entries from different arrival directions and two complete circuits; one timed and one distance-based hold when available.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Visual transition and circling decision · 18 minutes

Teach / coach: Brief approach category, published circling area, wind, visual references, missed procedure, and aircraft/school operating limits. Demonstrate the ground track and decision gates on a chart before using an authorized visual or device scenario.

Student exercise: Maintain required visual contact and circling altitude while positioning for a stabilized landing. Reject a poor position early; explain and fly the appropriate missed procedure if visual contact or a safe landing path is lost.

Practice: One straight-in visual transition and one authorized circling scenario; the device can rehearse loss of visual reference.

Look for: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Watch / correct: Tightening bank to rescue the landing - abandon the unstable position and go missed.

Watch / correct: Visual contact lost with no plan - brief the missed transition before starting the circle.

Stop / transition: Go missed immediately when required references, protected-area assurance, or a stabilized landing path are lost.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Visual transition and circling decision; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief - 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 36 - source page 59 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 37 · Preparation: Graphical Forecasts for Aviation, Terminal Aerodrome Forecasts, METARs

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 38 · Approach integration and manual reversion

[Return to lesson index](#)

120 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-32 Air 08-27	[] Autopilot modes and manual reversion	Select heading, navigation, altitude, and permitted climb/descent modes.
32-51 Air 27-46	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
51-70 Air 46-65	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
70-89 Air 65-84	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
89-107 Air 84-102	[] Visual transition and circling decision	Maintain required visual contact and circling altitude while positioning for a stabilized landing.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 38

Practice approach integration and manual reversion. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Autopilot modes and manual reversion · 19 minutes

Teach / coach: Brief the approved engagement envelope, limitations, disconnect methods, and failure signs. Demonstrate each installed mode by stating what is armed, active, commanded, and actually happening; then disconnect and hand-fly the same segment.

Student exercise: Select heading, navigation, altitude, and permitted climb/descent modes. Confirm annunciations and resulting flight path after every change. Recover from a simulated wrong mode or failed automation using the approved disconnect and manual control.

Practice: One mode sequence, one capture verification, and two different disconnect/reversion methods if installed.

Look for: Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.

Watch / correct: Armed mistaken for active - point to annunciations and confirm actual capture.

Watch / correct: Fighting engaged servos - use the approved disconnect and stabilize manually.

Stop / transition: Disengage for unexpected flight path, unresolved mode confusion, or operation outside approved limitations.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Autopilot modes and manual reversion; current ACS and applicable POH/AFM

Failure recognition and backup control · 19 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 19 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Lateral approach and missed approach · 19 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

Visual transition and circling decision · 18 minutes

Teach / coach: Brief approach category, published circling area, wind, visual references, missed procedure, and aircraft/school operating limits. Demonstrate the ground track and decision gates on a chart before using an authorized visual or device scenario.

Student exercise: Maintain required visual contact and circling altitude while positioning for a stabilized landing. Reject a poor position early; explain and fly the appropriate missed procedure if visual contact or a safe landing path is lost.

Practice: One straight-in visual transition and one authorized circling scenario; the device can rehearse loss of visual reference.

Look for: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Watch / correct: Tightening bank to rescue the landing - abandon the unstable position and go missed.

Watch / correct: Visual contact lost with no plan - brief the missed transition before starting the circle.

Stop / transition: Go missed immediately when required references, protected-area assurance, or a stabilized landing path are lost.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Visual transition and circling decision; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief - 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Holding Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 38 - source page 61 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 39 · Stage II assessment

[Return to lesson index](#)

120 min check | Brief 39 | Preflight 15 | Debrief 21

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-32 Air 08-27	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
32-51 Air 27-46	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
51-70 Air 46-65	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
70-89 Air 65-84	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
89-107 Air 84-102	[] Visual transition and circling decision	Maintain required visual contact and circling altitude while positioning for a stabilized landing.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 39

Assess independent performance and decisions using current FAA criteria; pause coaching except to protect safety.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 39 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Assess the learner's departure, configuration, and course intercept independently; give no coaching during the check.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Clearance, holding entry, and wind correction · 19 minutes

Teach / coach: Assign the briefed clearance, holding entry, and wind correction scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 19 minutes

Teach / coach: Assign the briefed ils and approved vertical guidance scenario and observe setup, execution, and decisions without coaching. Assess this result: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Failure recognition and backup control - 19 minutes

Teach / coach: Assign the briefed failure recognition and backup control scenario and observe setup, execution, and decisions without coaching. Assess this result: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Lateral approach and missed approach - 19 minutes

Teach / coach: Assign the briefed lateral approach and missed approach scenario and observe setup, execution, and decisions without coaching. Assess this result: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

Visual transition and circling decision · 18 minutes

Teach / coach: Assign the briefed visual transition and circling decision scenario and observe setup, execution, and decisions without coaching. Assess this result: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Maintain required visual contact and circling altitude while positioning for a stabilized landing. Reject a poor position early; explain and fly the appropriate missed procedure if visual contact or a safe landing path is lost.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Watch / correct: Tightening bank to rescue the landing - abandon the unstable position and go missed.

Watch / correct: Visual contact lost with no plan - brief the missed transition before starting the circle.

Stop / transition: Go missed immediately when required references, protected-area assurance, or a stabilized landing path are lost.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Visual transition and circling decision; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 21 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: Departure Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Standard Terminal Arrival Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures with an Autopilot. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 39 - source page 63, 64 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

IFR cross-country and readiness

Apply the instrument skills to routes, changing conditions, contingencies, and independent assessments.

Lesson 40 - Preparation: Chart Supplements, IFR Preferred Routes, En Route Low Altitude IFR Chart

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation](#); assignments on the referenced TCO pages

[FAA aviation handbooks](#); use aircraft/equipment supplements too

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 41 · Preparation: Charts and Publications, Weather Briefing, NOTAMs

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 42 · IFR cross-country rehearsal

[Return to lesson index](#)

120 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-35 Device	[] IFR route, weather, and contingency decisions	File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg.
35-62 Device	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
62-89 Device	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
89-115 Device	[] Visual transition and circling decision	Maintain required visual contact and circling altitude while positioning for a stabilized landing.
115-120 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 42

Practice ifr cross-country rehearsal. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

IFR route, weather, and contingency decisions · 27 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg. Select an alternate before the escape option disappears; explain the applicable lost-communications response and request assistance while communications work.

Practice: At least two route checkpoints, one changed-clearance readback, one diversion decision, and approaches at planned destinations.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

Lateral approach and missed approach · 27 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 27 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Visual transition and circling decision · 26 minutes

Teach / coach: Brief approach category, published circling area, wind, visual references, missed procedure, and aircraft/school operating limits. Demonstrate the ground track and decision gates on a chart before using an authorized visual or device scenario.

Student exercise: Maintain required visual contact and circling altitude while positioning for a stabilized landing. Reject a poor position early; explain and fly the appropriate missed procedure if visual contact or a safe landing path is lost.

Practice: One straight-in visual transition and one authorized circling scenario; the device can rehearse loss of visual reference.

Look for: Keeps the aircraft within the protected area, preserves required visibility, and makes an early conservative missed decision.

Watch / correct: Tightening bank to rescue the landing - abandon the unstable position and go missed.

Watch / correct: Visual contact lost with no plan - brief the missed transition before starting the circle.

Stop / transition: Go missed immediately when required references, protected-area assurance, or a stabilized landing path are lost.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Visual transition and circling decision; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 42 - source page 67 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 43 · Multi-airport IFR cross-country

[Return to lesson index](#)

180 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-90 Air 08-85	[] IFR route, weather, and contingency decisions	File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg.
90-129 Air 85-124	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
129-167 Air 124-162	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
167-177 Air 162-172	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
177-180 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 43

Practice multi-airport ifr cross-country. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.
- Source scenario uses multiple airports including one beyond 75 NM. Treat this as a source exercise target; use the separate current Part 61 cross-country requirements for certificate eligibility.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

IFR route, weather, and contingency decisions · 77 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg. Select an alternate before the escape option disappears; explain the applicable lost-communications response and request assistance while communications work.

Practice: At least two route checkpoints, one changed-clearance readback, one diversion decision, and approaches at planned destinations.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

Lateral approach and missed approach · 39 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 38 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from Straight-In / Circling Approaches. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 43 - source page 68 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 44 · Weather and degraded route rehearsal

[Return to lesson index](#)

120 min device | Brief 16 | Preflight 10 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Device	[] Device setup and clearance review	Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.
08-30 Device	[] IFR route, weather, and contingency decisions	File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg.
30-52 Device	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
52-73 Device	[] DME arc and position cross-check	Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead...
73-94 Device	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
94-115 Device	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
115-120 Device	[] Secure device and verify results	Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 44

Practice weather and degraded route rehearsal. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.
- Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Device setup and clearance review · 8 minutes

Teach / coach: Verify the device configuration, intended training credit, database and weather scenario. Set failures only by the approved device method; state whether pauses are instruction or evaluation.

Student exercise: Set up the cockpit as in the aircraft, verify navigation sources, copy the starting clearance, and explain today's equipment/failure and approach plans.

Practice: One complete setup and clearance readback.

Look for: Can explain the actual simulated aircraft/equipment state before the scenario runs.

Watch / correct: Unverified setup - compare the device state with the intended aircraft/manual.

Stop / transition: Do not log unapproved credit or continue with an invalid configuration.

IFR route, weather, and contingency decisions · 22 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: File the plan when appropriate, verify the clearance, fly the route and altitude restrictions, and update ETA/fuel/weather at each leg. Select an alternate before the escape option disappears; explain the applicable lost-communications response and request assistance while communications work.

Practice: At least two route checkpoints, one changed-clearance readback, one diversion decision, and approaches at planned destinations.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

Failure recognition and backup control · 22 minutes

Teach / coach: Brief a realistic failure supported by the aircraft or device. Reveal its indications, require independent confirmation, then demonstrate the remaining scan. Never disable equipment that compromises the actual flight or required equipment.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: Two different failures and one heading change each direction; device sessions may freeze only after the attempt for explanation.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

DME arc and position cross-check · 21 minutes

Teach / coach: Brief the charted arc, altitude, direction, lead points, and approved distance source.

Demonstrate a small heading change to maintain distance and anticipation of the inbound course; explain slant-range and equipment substitution limitations.

Student exercise: Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead the inbound-course intercept.

Practice: One arc each direction in the device; aircraft session one published arc plus a repeat of the intercept if time permits.

Look for: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.

Watch / correct: Late arc intercept - calculate/brief lead using speed and geometry before the turn.

Watch / correct: Large distance chase - make small heading changes and reassess trend.

Stop / transition: Ask for vectors or another procedure if equipment, clearance, or containment is uncertain.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; DME arc and position cross-check; current ACS and applicable POH/AFM

Lateral approach and missed approach · 21 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One complete approach and one vectored approach; at least one missed approach.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance · 21 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: Two approaches with different guidance where available: one landing transition and one missed approach.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Secure device and verify results · 5 minutes

Teach / coach: Stop the scenario after a stable endpoint, review objective evidence, and distinguish simulated/device time from aircraft or airborne time.

Student exercise: Complete the simulated shutdown/checklist, describe the main error and correction, and identify the remaining practice task.

Practice: One complete scenario wrap-up.

Look for: Ends with an accurate account of what was practiced and what remains.

Watch / correct: Treating paused time as flight - record only credit permitted by the device authorization.

Stop / transition: End before fatigue makes further practice ineffective.

Reference: Applicable POH/AFM and checklist; FAA Instrument Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from Straight-In / Circling Approaches. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 44 - source page 69 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 45 · Long IFR cross-country

[Return to lesson index](#)

240 min dual | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-34 Air 08-29	[] Autopilot modes and manual reversion	Select heading, navigation, altitude, and permitted climb/descent modes.
34-63 Air 29-58	[] IFR route leg 1: navigation and decisions	Fly cleared route leg 1 to the airport assigned to approach A.
63-106 Air 58-101	[] A: selected lateral procedure	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
106-135 Air 101-130	[] IFR route leg 2: navigation and decisions	Fly cleared route leg 2 to the airport assigned to approach B.
135-178 Air 130-173	[] B: selected ILS procedure	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
178-206 Air 173-201	[] IFR route leg 3: navigation and decisions	Fly cleared route leg 3 to the airport assigned to approach C.
206-227 Air 201-222	[] C: selected RNAV procedure	Fly the third planned RNAV approach at its assigned airport.
227-237 Air 222-232	[] Destination C: qualifying landing	After approach C, make the legal visual transition and a stabilized landing at planned destination C, more than 50 NM straight-line from the original departure...
237-240 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 45

Practice long ifr cross-country. Move from explanation and demonstration to an independent attempt, with time reserved for a useful correction.

Source clarification: The lesson heading prints Stage II; the course section/sequence places this lesson in Stage III. Navigation uses the surrounding course section.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMs, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.
- Before dispatch choose and name approach A, B, and C at the actual planned airports; the instructor verifies three different qualifying kinds using navigation systems (for example VOR, ILS, and an eligible RNAV procedure). Assign the lateral, ILS, and RNAV blocks below to those procedures. Arrange actual clearances and equipment eligibility; substitution must preserve the regulatory requirement or be recorded as nonqualifying.
- End this qualifying plan at destination C, with an actual landing more than 50 NM straight-line from the original departure point under §61.1. Verify that destination and landing before dispatch. A later homeward flight is a separate leg/session; do not replace this landing with three misses and a home landing.
- For Part 61 §61.65(d)(2)(ii) credit: plan the required 250 NM along airways or ATC-directed routing, under IFR with a filed flight plan, an approach at each airport, and three different kinds of approaches using navigation systems. Verify/log the actual qualifying flight separately; the source Part 141 routing note alone is insufficient.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.

- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Monitor actual conditions and model a concise post-takeoff configuration flow.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Autopilot modes and manual reversion · 21 minutes

Teach / coach: Brief the approved engagement envelope, limitations, disconnect methods, and failure signs. Demonstrate each installed mode by stating what is armed, active, commanded, and actually happening; then disconnect and hand-fly the same segment.

Student exercise: Select heading, navigation, altitude, and permitted climb/descent modes. Confirm annunciations and resulting flight path after every change. Recover from a simulated wrong mode or failed automation using the approved disconnect and manual control.

Practice: One mode sequence, one capture verification, and two different disconnect/reversion methods if installed.

Look for: Monitors rather than assumes automation behavior; disconnects early when it no longer supports the plan.

Watch / correct: Armed mistaken for active - point to annunciations and confirm actual capture.

Watch / correct: Fighting engaged servos - use the approved disconnect and stabilize manually.

Stop / transition: Disengage for unexpected flight path, unresolved mode confusion, or operation outside approved limitations.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Autopilot modes and manual reversion; current ACS and applicable POH/AFM

IFR route leg 1: navigation and decisions · 29 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: Fly cleared route leg 1 to the airport assigned to approach A. Verify route/altitude restrictions and navigation source; update fuel, weather and alternate options. Complete the required approach at that airport before continuing under the next clearance.

Practice: Verify cleared route/altitude, ETA, fuel and current destination conditions on this leg.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

A: selected lateral procedure · 43 minutes

Teach / coach: Brief procedure eligibility, source, course, altitudes, MDA, missed point, visual requirements, and missed route. Demonstrate a stabilized descent using approved advisory vertical guidance only as advisory; distinguish LNAV/LP from vertically guided minima.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report. Fly the briefed missed approach and cleared onward route unless an actual landing is planned. If landing, allocate additional taxi/departure movement time and extend the plan; parked stop time remains separate.

Practice: Fly the named approach at its planned airport and record actual procedure/navigation system; complete a legal landing or missed as planned. The actual flown route must include an approach at each airport.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

IFR route leg 2: navigation and decisions · 29 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: Fly cleared route leg 2 to the airport assigned to approach B. Verify route/altitude restrictions and navigation source; update fuel, weather and alternate options. Complete the required approach at that airport before continuing under the next clearance.

Practice: Verify cleared route/altitude, ETA, fuel and current destination conditions on this leg.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

B: selected ILS procedure · 43 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent. Fly the briefed missed approach and cleared onward route unless an actual landing is planned. If landing, allocate additional taxi/departure movement time and extend the plan; parked stop time remains separate.

Practice: Fly the named approach at its planned airport and record actual procedure/navigation system; complete a legal landing or missed as planned. The actual flown route must include an approach at each airport.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

IFR route leg 3: navigation and decisions · 28 minutes

Teach / coach: Build an actual route with current weather, NOTAMs, alternates, fuel, departure/arrival restrictions, and approach options. In flight present one credible weather or routing change and require a decision supported by fuel and aircraft capability; rehearse lost communications without stopping real radio use.

Student exercise: Fly cleared route leg 3 to the airport assigned to approach C. Verify route/altitude restrictions and navigation source; update fuel, weather and alternate options. Complete the required approach at that airport before continuing under the next clearance.

Practice: Verify cleared route/altitude, ETA, fuel and current destination conditions on this leg.

Look for: Remains ahead of route and weather, keeps reserves and terrain/airspace protection, and communicates changes promptly.

Watch / correct: Continuing toward deteriorating weather - identify the last safe alternate and decide before reaching it.

Watch / correct: Clearance lost in the flight plan - compare active route against the actual amended clearance.

Stop / transition: Real weather, aircraft capability, or fuel margin controls the route; terminate a training scenario before it reduces safe options.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; IFR route, weather, and contingency decisions; current ACS and applicable POH/AFM

C: selected RNAV procedure · 21 minutes

Teach / coach: Contrast ILS, LPV, and LNAV/VNAV guidance, minima, equipment requirements, and failure responses. LP provides lateral guidance only; it is not an APV. For test credit, consult current ACS conditions for the selected LPV procedure rather than labeling every LPV a precision approach.

Student exercise: Fly the third planned RNAV approach at its assigned airport. Verify equipment and integrity, the actual guidance annunciation, selected chart minima, and legal visual transition or missed approach. Choose an approach kind that the instructor has confirmed meets the three-kind Part 61 plan; LPV credit/classification must be checked against the applicable standard.

Practice: Fly the named approach at its planned airport and record actual procedure/navigation system; complete a legal landing or missed as planned. The actual flown route must include an approach at each airport.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; RNAV approach and vertical-guidance eligibility; current ACS and applicable POH/AFM

Destination C: qualifying landing · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: After approach C, make the legal visual transition and a stabilized landing at planned destination C, more than 50 NM straight-line from the original departure point. Go missed whenever required for safety; if the qualifying landing cannot be made, use a safe alternate and treat the original training-credit plan as unmet. The flight ends and is secured at this destination.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief · 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from Straight-In / Circling Approaches. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 45 - source page 70 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 46 · Preparation: Instrument Rating Knowledge Test Review, Pilot Qualifications, Weather Information

Prepare the knowledge and cockpit decisions needed for the next flight. Use the source pages and corresponding course assignment before meeting the instructor.

[Return to lesson index](#)

Ground preparation objectives

- Explain the source preparation topics and identify their application to the next flight.
- Bring unresolved questions and aircraft-specific differences to the briefing.

Preparation

- Use the corresponding current course ground lesson and referenced source pages.

[Corresponding Sporty's course preparation; assignments on the referenced TCO pages](#)

[FAA aviation handbooks; use aircraft/equipment supplements too](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 46a · Independent end-of-course review

[Return to lesson index](#)

120 min check | Brief 39 | Preflight 15 | Debrief 21

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-29 Air 08-24	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
29-45 Air 24-40	[] Recognize and recover unusual attitudes	Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation...
45-61 Air 40-56	[] DME arc and position cross-check	Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead...
61-77 Air 56-72	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
77-92 Air 72-87	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
92-107 Air 87-102	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 46a

Assess independent performance and decisions using current FAA criteria; pause coaching except to protect safety.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.
- Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 39 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Assess the learner's departure, configuration, and course intercept independently; give no coaching during the check.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Clearance, holding entry, and wind correction · 16 minutes

Teach / coach: Assign the briefed clearance, holding entry, and wind correction scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Recognize and recover unusual attitudes · 16 minutes

Teach / coach: Assign the briefed recognize and recover unusual attitudes scenario and observe setup, execution, and decisions without coaching. Assess this result: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation or automation confusion caused the upset.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Watch / correct: Control response before diagnosis - identify the attitude using reliable cues first.

Watch / correct: Recovery creates a secondary stall/overspeed - manage angle of attack, bank, power, and load according to the approved procedure.

Stop / transition: Instructor ends setup or takes control before aircraft limits, safety altitude, or recovery margin are threatened.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Recognize and recover unusual attitudes; current ACS and applicable POH/AFM

DME arc and position cross-check · 16 minutes

Teach / coach: Assign the briefed dme arc and position cross-check scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead the inbound-course intercept.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.

Watch / correct: Late arc intercept - calculate/brief lead using speed and geometry before the turn.

Watch / correct: Large distance chase - make small heading changes and reassess trend.

Stop / transition: Ask for vectors or another procedure if equipment, clearance, or containment is uncertain.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; DME arc and position cross-check; current ACS and applicable POH/AFM

Lateral approach and missed approach - 16 minutes

Teach / coach: Assign the briefed lateral approach and missed approach scenario and observe setup, execution, and decisions without coaching. Assess this result: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance - 15 minutes

Teach / coach: Assign the briefed ils and approved vertical guidance scenario and observe setup, execution, and decisions without coaching. Assess this result: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Failure recognition and backup control · 15 minutes

Teach / coach: Assign the briefed failure recognition and backup control scenario and observe setup, execution, and decisions without coaching. Assess this result: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 21 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- The source lists more tasks than can receive full teaching and multiple repetitions in this target. Today's named exercise blocks are the priority; select additional reviews from the source only when time and proficiency permit, and carry unmet tasks forward.
- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Instrument Flight. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Departure, En route, and Arrival Operations. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from Straight-In/ Circling Approach. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Autopilot Operations. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 46a - source page 72 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 47 · Final proficiency rehearsal

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120 min check | Brief 16 | Preflight 15 | Debrief 10

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-29 Air 08-24	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
29-45 Air 24-40	[] Recognize and recover unusual attitudes	Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation...
45-61 Air 40-56	[] DME arc and position cross-check	Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead...
61-77 Air 56-72	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
77-92 Air 72-87	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
92-107 Air 87-102	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 47

Assess independent performance and decisions using current FAA criteria; pause coaching except to protect safety.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.
- Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 16 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Assess the learner's departure, configuration, and course intercept independently; give no coaching during the check.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Clearance, holding entry, and wind correction · 16 minutes

Teach / coach: Assign the briefed clearance, holding entry, and wind correction scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Recognize and recover unusual attitudes · 16 minutes

Teach / coach: Assign the briefed recognize and recover unusual attitudes scenario and observe setup, execution, and decisions without coaching. Assess this result: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation or automation confusion caused the upset.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Watch / correct: Control response before diagnosis - identify the attitude using reliable cues first.

Watch / correct: Recovery creates a secondary stall/overspeed - manage angle of attack, bank, power, and load according to the approved procedure.

Stop / transition: Instructor ends setup or takes control before aircraft limits, safety altitude, or recovery margin are threatened.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Recognize and recover unusual attitudes; current ACS and applicable POH/AFM

DME arc and position cross-check · 16 minutes

Teach / coach: Assign the briefed dme arc and position cross-check scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead the inbound-course intercept.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.

Watch / correct: Late arc intercept - calculate/brief lead using speed and geometry before the turn.

Watch / correct: Large distance chase - make small heading changes and reassess trend.

Stop / transition: Ask for vectors or another procedure if equipment, clearance, or containment is uncertain.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; DME arc and position cross-check; current ACS and applicable POH/AFM

Lateral approach and missed approach - 16 minutes

Teach / coach: Assign the briefed lateral approach and missed approach scenario and observe setup, execution, and decisions without coaching. Assess this result: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance - 15 minutes

Teach / coach: Assign the briefed ils and approved vertical guidance scenario and observe setup, execution, and decisions without coaching. Assess this result: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Failure recognition and backup control · 15 minutes

Teach / coach: Assign the briefed failure recognition and backup control scenario and observe setup, execution, and decisions without coaching. Assess this result: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief - 10 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Instrument Flight. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Departure, En route, and Arrival Operations. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from Straight-In/ Circling Approach. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Autopilot Operations. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 47 - source page 73 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard

Lesson 48 · End-of-course assessment

[Return to lesson index](#)

120 min check | Brief 39 | Preflight 15 | Debrief 21

Publisher lesson target; SUAREZ.CFI phase allocation is recommended. Brief/debrief are separate from flight.

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-05 Ground	[] Start, taxi, and departure checks	Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before...
05-13 Air 00-08	[] Takeoff, climb, and intercept	Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course.
13-29 Air 08-24	[] Clearance, holding entry, and wind correction	Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind.
29-45 Air 24-40	[] Recognize and recover unusual attitudes	Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation...
45-61 Air 40-56	[] DME arc and position cross-check	Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead...
61-77 Air 56-72	[] Lateral approach and missed approach	Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or...
77-92 Air 72-87	[] ILS and approved vertical guidance	Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized...
92-107 Air 87-102	[] Failure recognition and backup control	Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources.
107-117 Air 102-112	[] Return, landing, and go-around decision	Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved...
117-120 Ground	[] Taxi in, shutdown, and aircraft checks	Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 48

Assess independent performance and decisions using current FAA criteria; pause coaching except to protect safety.

Aircraft-neutral: set speeds, configurations, limitations, approved simulation methods, and any endorsement conditions from the actual aircraft and current rules.

Objectives

- Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.
- Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.
- Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.
- Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.
- Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.
- Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Before the flight

- Complete the corresponding ground preparation and review the referenced source pages.
- Before aircraft flight: confirm weather/NOTAMS, route/airspace, weight/balance, performance, aircraft airworthiness, personal readiness, and stop/go-around gates.
- Before simulated instrument work: agree who maintains outside lookout and how control transfer and view-limiting-device removal work.

Briefing · 39 minutes

- Agree on the named exercise priorities, aircraft-specific procedures, stop/takeover gates, weather/fuel/route limits, and how remaining source tasks will be carried forward.
- Learner explains the initial plan and one credible failure/alternate decision. Instructor corrects misunderstanding before departure.

Start, taxi, and departure checks · 5 minutes

Teach / coach: Coach checklist discipline and runway verification; during checks observe independently and intervene only for safety.

Student exercise: Start and taxi using the approved checklist, review hotspots, read back restrictions, hold short correctly, and complete run-up/departure checks before entering the runway.

Practice: One complete ground departure flow.

Look for: Positive runway/clearance verification, deliberate checklist and configuration checks.

Watch / correct: Heads-down taxi - stop before configuring or troubleshooting.

Stop / transition: Stop taxi for uncertain clearance, traffic, or an unresolved abnormality.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Takeoff, climb, and intercept · 8 minutes

Teach / coach: Brief the approved takeoff/abort and climb procedure. Assess the learner's departure, configuration, and course intercept independently; give no coaching during the check.

Student exercise: Verify runway and clearance, take off using the approved technique, establish climb, configure/check, and intercept the cleared or planned course. A view-limiting device is used only after the outside-lookout/control arrangement is established.

Practice: One departure and course intercept.

Look for: Maintains control, follows the route/clearance, and completes checks at an appropriate workload.

Watch / correct: Fixation on route programming - stabilize flight and defer the entry.

Stop / transition: Abort or return under the actual aircraft procedure and conditions; no universal return altitude.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Clearance, holding entry, and wind correction · 16 minutes

Teach / coach: Assign the briefed clearance, holding entry, and wind correction scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Copy and verify the clearance, select a defensible entry, cross the fix, establish the hold, and adjust outbound timing/distance and heading for wind. Communicate any inability and track clearance limits.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains within the protected procedure, meets altitude/speed constraints, and corrects timing without losing control.

Watch / correct: Entry guessed from a moving display - orient the published inbound course and aircraft arrival direction before choosing.

Watch / correct: Timing starts at the wrong place - identify the published timing convention and fix crossing.

Stop / transition: Request clarification or vectors before reaching an unclear limit; abandon exercise for unsafe control or containment.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Clearance, holding entry, and wind correction; current ACS and applicable POH/AFM

Recognize and recover unusual attitudes · 16 minutes

Teach / coach: Assign the briefed recognize and recover unusual attitudes scenario and observe setup, execution, and decisions without coaching. Assess this result: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Diagnose the attitude from reliable indications, use the aircraft-appropriate recovery priorities, stabilize attitude/power/airspeed, and explain how fixation or automation confusion caused the upset.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Recognizes the condition, promptly reduces the immediate risk, avoids secondary loss of control, and re-establishes a useful scan.

Watch / correct: Control response before diagnosis - identify the attitude using reliable cues first.

Watch / correct: Recovery creates a secondary stall/overspeed - manage angle of attack, bank, power, and load according to the approved procedure.

Stop / transition: Instructor ends setup or takes control before aircraft limits, safety altitude, or recovery margin are threatened.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Recognize and recover unusual attitudes; current ACS and applicable POH/AFM

DME arc and position cross-check · 16 minutes

Teach / coach: Assign the briefed dme arc and position cross-check scenario and observe setup, execution, and decisions without coaching. Assess this result: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the station or approved substitute, intercept the arc, keep charted distance with incremental corrections, comply with altitude restrictions, and lead the inbound-course intercept.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Remains oriented to the fix, anticipates intercepts, and uses the authorized distance source correctly.

Watch / correct: Late arc intercept - calculate/brief lead using speed and geometry before the turn.

Watch / correct: Large distance chase - make small heading changes and reassess trend.

Stop / transition: Ask for vectors or another procedure if equipment, clearance, or containment is uncertain.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; DME arc and position cross-check; current ACS and applicable POH/AFM

Lateral approach and missed approach - 16 minutes

Teach / coach: Assign the briefed lateral approach and missed approach scenario and observe setup, execution, and decisions without coaching. Assess this result: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Load or tune and identify the procedure, brief it, fly the cleared entry, meet restrictions, descend to the published MDA, and hold it until a legal landing or missed approach. At the missed point, execute the aircraft procedure, navigation sequence, and ATC report.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Maintains lateral guidance and altitude restrictions, applies the correct minima, and promptly flies a missed approach when required.

Watch / correct: Advisory glidepath treated as authorization below MDA - use the selected chart minima and required visual conditions.

Watch / correct: Early descent after a shortcut - verify the cleared segment and remaining altitude restrictions.

Stop / transition: Go missed when required by the clearance/procedure, guidance loss, unstable approach, or unmet landing conditions.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Lateral approach and missed approach; current ACS and applicable POH/AFM

ILS and approved vertical guidance - 15 minutes

Teach / coach: Assign the briefed ils and approved vertical guidance scenario and observe setup, execution, and decisions without coaching. Assess this result: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Brief an ILS or approved vertically guided procedure, verify the active navigation source and annunciation, intercept as cleared, establish a stabilized configuration, and follow lateral/vertical guidance to the selected DA. Execute a missed approach when the required visual conditions are absent.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Uses the correct source, minima, and annunciations, stabilizes early, and makes a timely decision at DA.

Watch / correct: Wrong CDI source - verify the source before intercept and compare with the chart.

Watch / correct: Below-path correction by steep descent - recover a stable path or go missed; do not chase near the ground.

Stop / transition: Go missed for lost required guidance, instability, unavailable visual conditions, or an instruction requiring it.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; ILS and approved vertical guidance; current ACS and applicable POH/AFM

Failure recognition and backup control · 15 minutes

Teach / coach: Assign the briefed failure recognition and backup control scenario and observe setup, execution, and decisions without coaching. Assess this result: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability. Record errors for the parked debrief; intervene immediately for safety. New instruction and corrective demonstration belong in a separate training flight.

Student exercise: Identify the simulated failed instrument, cross-check its reliability, establish control with valid indications, and use backup attitude/heading resources. Make timed turns and compass-supported turns with appropriate acceleration/dip error awareness.

Practice: One independent demonstration of each selected task; remedial coaching belongs in a later training flight.

Look for: Rejects misleading data, controls the aircraft with valid sources, and communicates the degraded capability.

Watch / correct: Following a failed display - cross-check independent indications and isolate the bad source.

Watch / correct: Treating compass as steady during a maneuver - stabilize before reading and use timed/backup methods as appropriate.

Stop / transition: End the simulation for loss of orientation, unsafe deviations, unavailable reliable backup, or real equipment abnormality.

Reference: FAA Instrument Flying Handbook / Instrument Procedures Handbook; Failure recognition and backup control; current ACS and applicable POH/AFM

Return, landing, and go-around decision · 10 minutes

Teach / coach: Reassess fuel, weather, traffic, runway and learner workload before returning. Let the learner plan the arrival; avoid rescuing a late approach with last-minute prompts.

Student exercise: Obtain the arrival information/clearance, brief the runway and landing configuration, fly a stabilized approach, land if safe or go around using the approved procedure, then clear the runway.

Practice: One stable arrival and final landing; additional approaches are only those budgeted in exercise blocks.

Look for: Timely configuration and a decisive landing/go-around choice.

Watch / correct: Late arrival briefing - finish the plan before descent or request delay.

Stop / transition: Go around whenever the safe landing conditions are not met.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Taxi in, shutdown, and aircraft checks · 3 minutes

Teach / coach: Keep the learner focused on taxi and securing the airplane; gather teaching notes after movement stops.

Student exercise: Clear and identify the taxi route, perform after-landing checks when safe, park, shut down, secure, and report any equipment discrepancy.

Practice: One complete shutdown and securing flow.

Look for: Safe taxi and accurate checklist/equipment follow-up.

Watch / correct: Debrief during a hotspot - postpone discussion until parked.

Stop / transition: Stop for uncertain route or traffic; complete abnormal follow-up before another departure.

Reference: Applicable POH/AFM and checklist; FAA Airplane Flying Handbook

Completion criteria

- Maintains aircraft control, checklist/clearance discipline, and safe decisions throughout the complete session.
- Demonstrates the selected objectives at today's expected level: coached learning during introduction, fewer cues during review, and current ACS criteria independently at checks.
- Unmet or unobserved source tasks are identified for a later flight; elapsed time and checkmarks are not a proficiency endorsement.

Debrief • 21 minutes

- Learner names the strongest result and the first decision/control error.
- Instructor ties one observed error to its cause, gives a specific correction, and confirms the next practice priority.
- Compare actual exercise/route time with this plan and verify any regulatory experience credit independently.

Repeat or carry forward

- The source lists more tasks than can receive full teaching and multiple repetitions in this target. Today's named exercise blocks are the priority; select additional reviews from the source only when time and proficiency permit, and carry unmet tasks forward.
- Source task for another focused block if not observed today: ATC Clearances and Clearance Compliance. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Instrument Flight. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Recovery from Unusual Flight Attitudes. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Departure, En route, and Arrival Operations. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Missed Approach Procedures. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Landing from Straight-In/ Circling Approach. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.
- Source task for another focused block if not observed today: Autopilot Operations. Select the applicable FAA/aircraft procedure and schedule adequate instruction; no completion is implied by this checklist.

Study and sources

Corresponding Sporty's course preparation; assignments on the referenced TCO pages

FAA aviation handbooks; use aircraft/equipment supplements too

Sporty's Academy Instrument TCO - lesson 48 - source page 75 - Sporty's Academy Instrument 141 TCO V18 / 06-17-2026

FAA ACS criteria and evaluator instructions - source page Relevant areas of operation - Current FAA linked standard